

Topic: Design of a Low-Cost Industrial IoT Laboratory Platform Using PLC, Microcontrollers, and Single-Board Computers

Objective:

The objective of this project is to design and implement an **IoT system** consisting of a **programmable logic controller (PLC)**, **microcontroller boards**, and a **single-board computer (SBC)**, all of which communicate using industrial communication protocols such as **Modbus TCP** and **MQTT**.

The proposed system serves as a **prototype platform for educational and training laboratory activities** in the areas of **industrial automation**, **home automation (HA)**, and **Industrial IoT (IIoT)**.

System Architecture and Hardware Components

This project focuses on the integration of the following hardware components:

- 1) a low-cost PLC (e.g., Siemens LOGO! 8.4),
- 2) MCU boards based on Espressif Systems ESP32 SoCs or their variants, and
- 3) an SBC such as Raspberry Pi Foundation Raspberry Pi 4 or 5.

In this project, the MCU-based hardware components serve the following purposes:

- to provide a performance and functionality comparison with the PLC,
- to implement protocol converters or communication gateways, and
- to interface different types of sensors/actuators.

The **SBC** will serve as a **local server and IoT gateway**, hosting services such as the **MQTT broker, database, Node-RED flows**, and **monitoring dashboards**.

Scope of Work (Provided as an example)

1. PLC Development

- 1.1 Study and configure the **Siemens LOGO! 8 PLC** hardware and networking features.
- 1.2 Develop PLC programs using **LOGO! Soft Comfort**.
- 1.3 Implement control logic using **Function Block Diagram (FBD)** and **Ladder Diagram (LD)**.
- 1.4 Enable and test **Modbus TCP and MQTT communication** on the PLC.

2. Microcontroller Development

- 2.1 Develop firmware for ESP32-based microcontroller boards.
- 2.2 Interface sensors and actuators (e.g., relays, level sensors, switches).
- 2.3 Implement MQTT clients and Modbus protocol converters.
- 2.4 Compare performance and functionality with the PLC platform.

Scope of Work (Provided as an example)

3. System Integration

- 3.1 Set up a system prototype consisting of the PLC, microcontroller nodes, an SBC, and required peripheral hardware.
- 3.2 Establish communication between devices using Modbus TCP and MQTT.
- 3.3 Design and verify the overall network architecture.

4. SBC Gateway and Server Configuration

- 4.1 Install and configure an Ubuntu-based operating system on the SBC.
- 4.2 Deploy IoT gateway services, including:
 - Eclipse Mosquitto (an open-source MQTT broker)
 - Node-RED for data flows, automation logic and user interface
 - Optional: InfluxDB for data logging (database)
 - Optional: Grafana for dashboards and visualization
- 4.3 Optionally evaluate a home automation platform such as **Home Assistant** and compare it with **Node-RED**–based solutions.

Scope of Work (Provided as an example)

5. Application Development

5.1 Implement demonstration use cases, such as:

- Smart home automation
- Water pump or water tank level control

5.2 Integrate sensors, actuators, and monitoring dashboards.

6. Testing and Evaluation

6.1 Test communication reliability and latency.

6.2 Measure system performance (response time, throughput, CPU usage).

6.3 Compare PLC and MCU solutions in terms of cost, complexity, and scalability.

6.4 Validate system stability under continuous operation.

7. Documentation and Reporting

Integration of PLCs into IoT Systems and Applications

PLC Platform: PLC: Siemens LOGO! 8 (v8.4)

- **Communication protocol support:** MQTT, Modbus TCP
- **Software:** LOGO! Soft Comfort
- **Programming approaches:** Function Block Diagram (FBD) or Ladder Diagram (LD)

Examples of Open-source Software for Automation

- **Node-RED:** A flow-based visual programming tool used for wiring together devices, APIs, and online services for rapid IoT application development.
- **Eclipse Mosquitto:** A lightweight MQTT broker that manages publish/subscribe messaging between IoT devices.
- **Grafana:** A data visualization and dashboard platform for monitoring, analyzing, and displaying real-time system data.
- **InfluxDB:** A time-series database optimized for storing and querying sensor and telemetry data.
- **Home Automation (HA):** A widely used open-source platform for building and managing home automation systems.

Background: SCADA

- **SCADA (Supervisory Control and Data Acquisition)** is an **industrial control system architecture** used to monitor, supervise, and control distributed processes and equipment through centralized software.
- It collects real-time data from field devices such as sensors, actuators, **Programmable Logic Controllers (PLCs)**, and **Remote Terminal Units (RTUs)**, enabling operators to visualize system status, log data, generate alarms, and issue supervisory control commands.
- A **Remote Terminal Unit (RTU)** is a microprocessor-based field device used in industrial control and SCADA systems to interface with sensors and actuators at remote or distributed locations.
- **RTUs** collect real-time data from field instruments, perform basic local control or preprocessing, and communicate this information to a central control system or SCADA station through wired or wireless communication networks.

Background: IoT-based SCADA

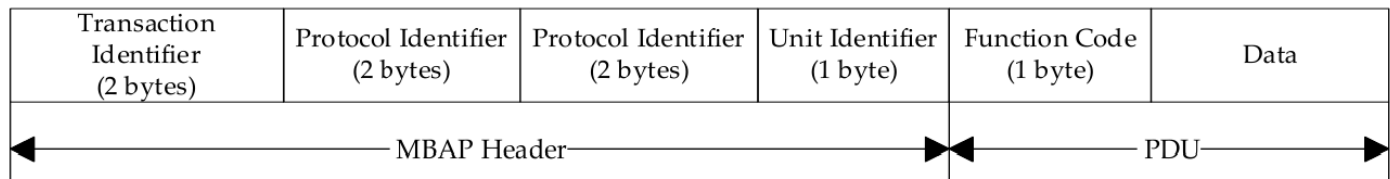
- **IoT-based SCADA**, also referred to as 4th-generation SCADA, extends traditional SCADA systems by integrating Internet of Things technologies, cloud computing, and open communication protocols.
- It uses IP-based networking and lightweight messaging protocols (e.g., **MQTT, HTTPS, REST APIs**) to connect distributed devices, gateways, and cloud platforms, enabling scalable, low-cost, and remote monitoring and control.
- Instead of proprietary SCADA hardware and software, **IoT-based SCADA** often employs edge devices, gateways, databases, and web dashboards running on standard computers or single-board computers.

Background: Modbus TCP (Modbus TCP/IP)

- **Modbus TCP** is an Ethernet-based industrial communication protocol that transmits Modbus messages over **TCP/IP networks**.
- It enables reliable **client-server communication** between industrial devices such as **PLCs**, remote terminal units (RTUs), sensors, and supervisory systems using standard Ethernet infrastructure.
- Unlike **serial Modbus (Modbus RS485)**, **Modbus TCP** encapsulates Modbus frames inside **TCP packets**, allowing higher data rates, longer distances, and easier integration with modern IT and IoT systems.
- **Modbus RTU** is a serial communication protocol that transmits Modbus messages in a compact binary format over serial interfaces such as **RS-485**.

Background: Modbus TCP Protocol

Modbus TCP is an industrial communication protocol that operates on TCP/IP and utilizes a Master-Slave architecture. The protocol's framework comprises the Modbus Application Protocol (MBAP Header), a 7-byte header, and the Protocol Data Unit (PDU) for data transmission.



Background: MQTT

- **MQTT (Message Queuing Telemetry Transport)** is a lightweight, **publish-subscribe messaging protocol** designed for reliable and efficient communication between distributed devices over TCP/IP networks.
- **MQTT** is widely used in Internet of Things (IoT), Industrial IoT (IIoT), and SCADA systems to exchange telemetry data between sensors, controllers, gateways, and cloud services with minimal bandwidth and power consumption.
- **MQTT** uses a **broker-based architecture**, where clients **publish** messages to **topics** and other clients **subscribe** to those topics to receive data.

- **Node-RED** is an **open-source, flow-based programming platform** designed for rapid development of IoT, industrial automation, and edge computing applications.
- It is based on **Node.js** and allows users to visually connect devices, protocols, APIs, and services using browser-based graphical flows rather than writing large amounts of traditional code.
- It is commonly **deployed on edge gateways, industrial PCs, or single-board computers** to act as middleware between field devices (PLCs, sensors, microcontrollers) and higher-level systems such as databases, dashboards, or cloud services.

Key Features of Node-RED

Flow-based visual programming

- Drag-and-drop editor in a web browser
- Logic built using connected “nodes”
- Faster prototyping and debugging

Built-in IIoT protocol support

- MQTT
- Modbus TCP/RTU
- HTTP/REST
- WebSocket
- TCP/UDP
- Serial communication

Dashboard and visualization

- Built-in web dashboard nodes
- Real-time charts, gauges, buttons, indicators
- Useful for SCADA-like monitoring

Easy integration and orchestration

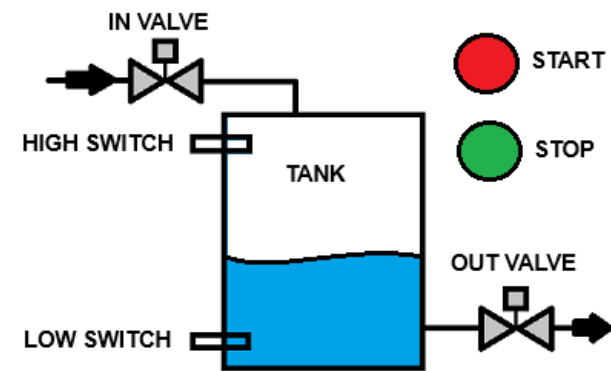
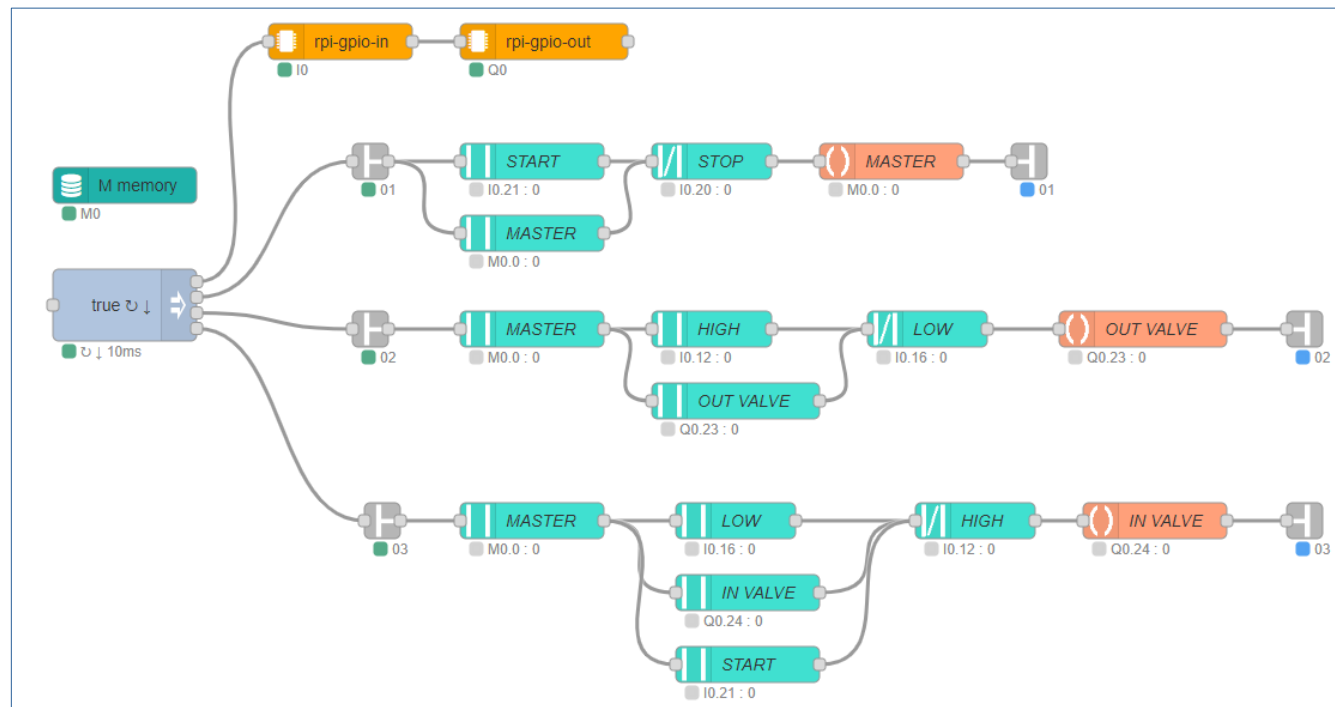
- Data routing and transformation between protocols
- Rule-based automation and workflows
- Acts as middleware between **OT (Operational Technology)** and **IT (Information Technology)** layers

Advantages of Using Node-RED in an IoT + PLC Project

Node-Red

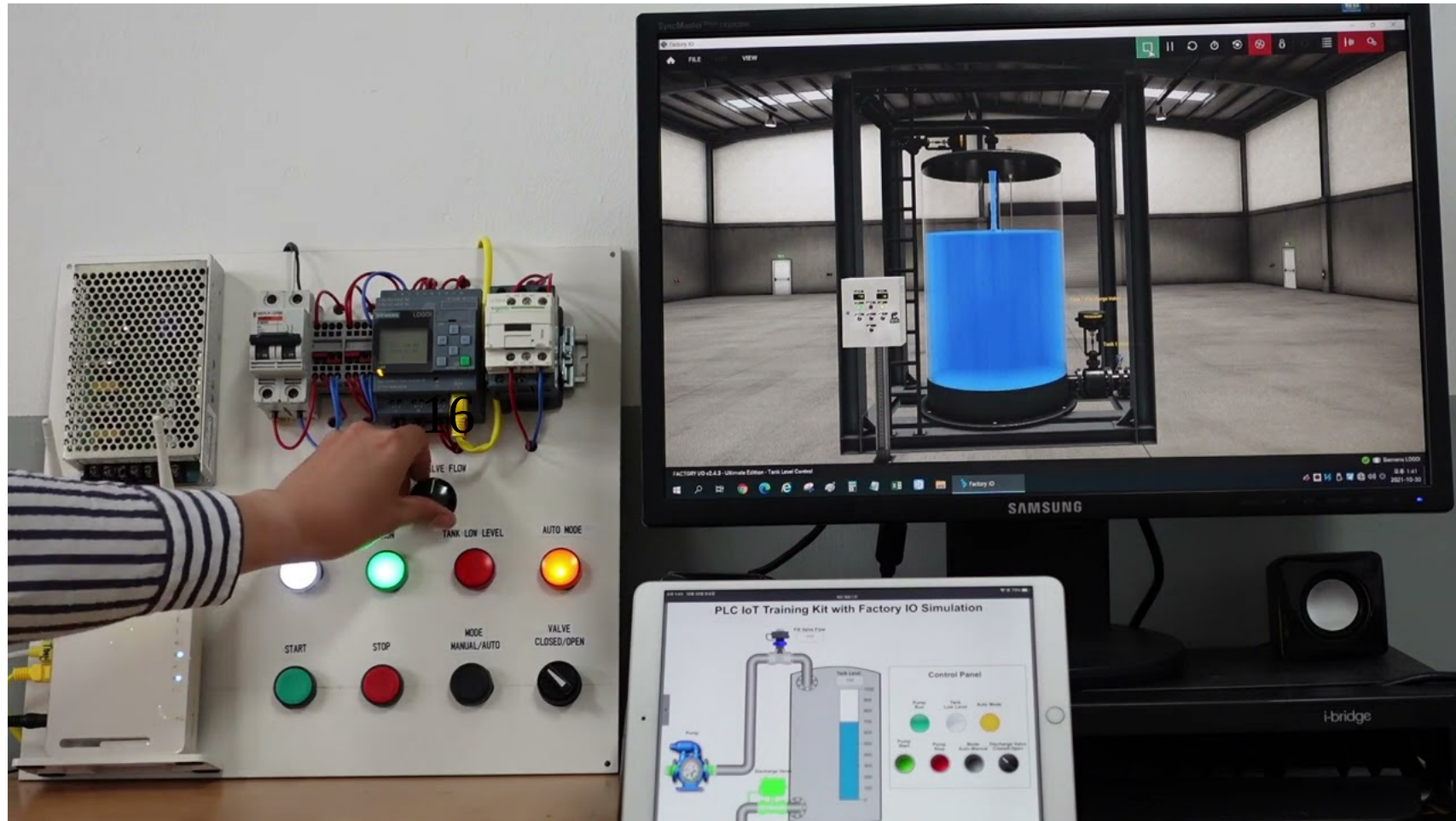
- Open source, platform-agnostic
- Rapid development with visual flows
- Support for industrial protocols such as:
 - Modbus (for PLC communication)
 - MQTT (for message distribution)
 - TCP/IP, OPC/OPC-UA (industrial communications)
- Support for database connections such as InfluxDB
- Built-in Dashboard UI components
- Can act as a **middleware IoT gateway**, bridging OT (PLC) with IT (cloud, analytics).
- Can be used to build a **low-cost, customizable SCADA-like system**.

Advantages of Using Node-RED in an IoT + PLC Project

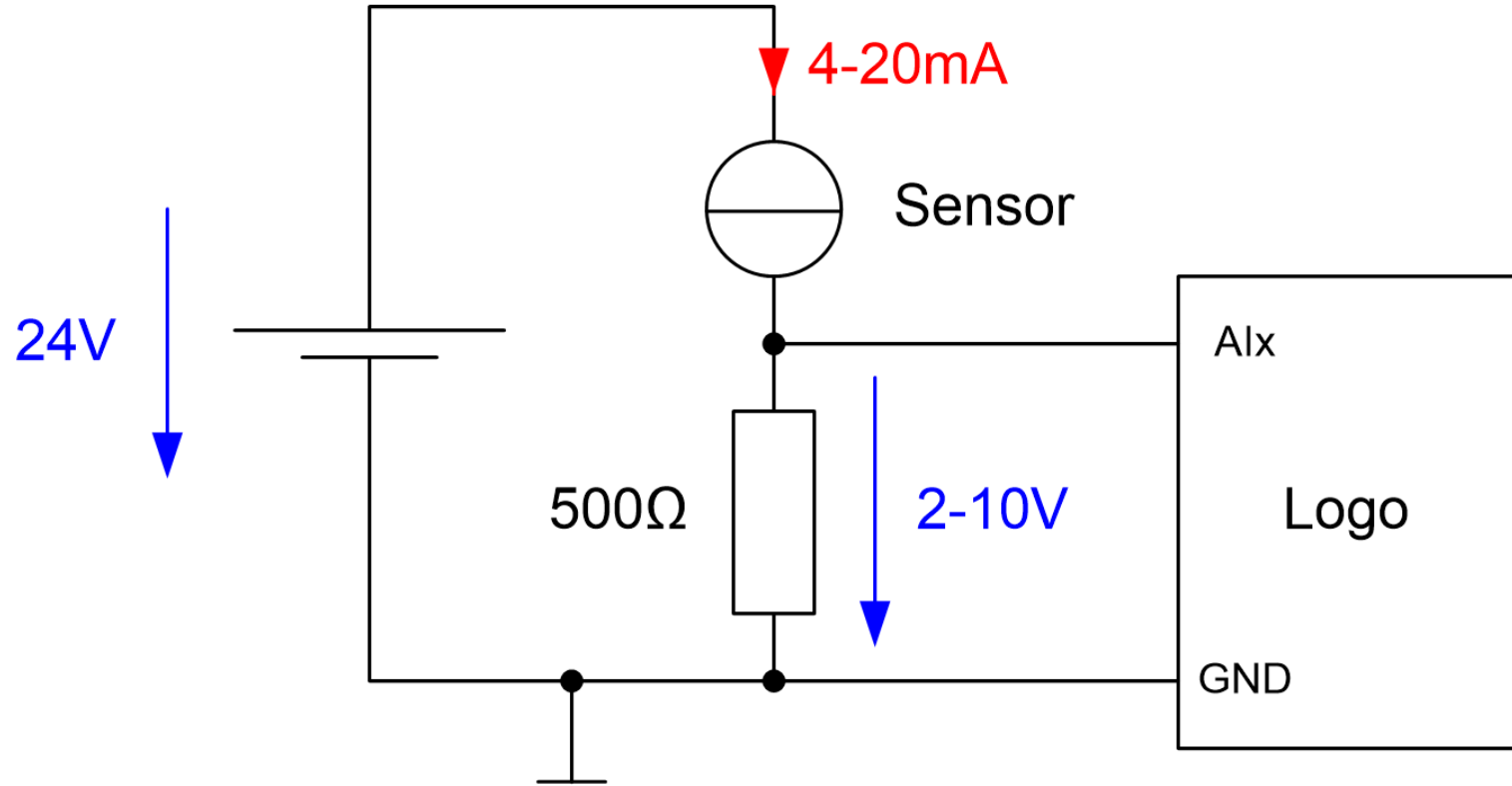


Source: <https://github.com/redplc/redplc>

Siemens Logo 8 for Tank Water Level Control



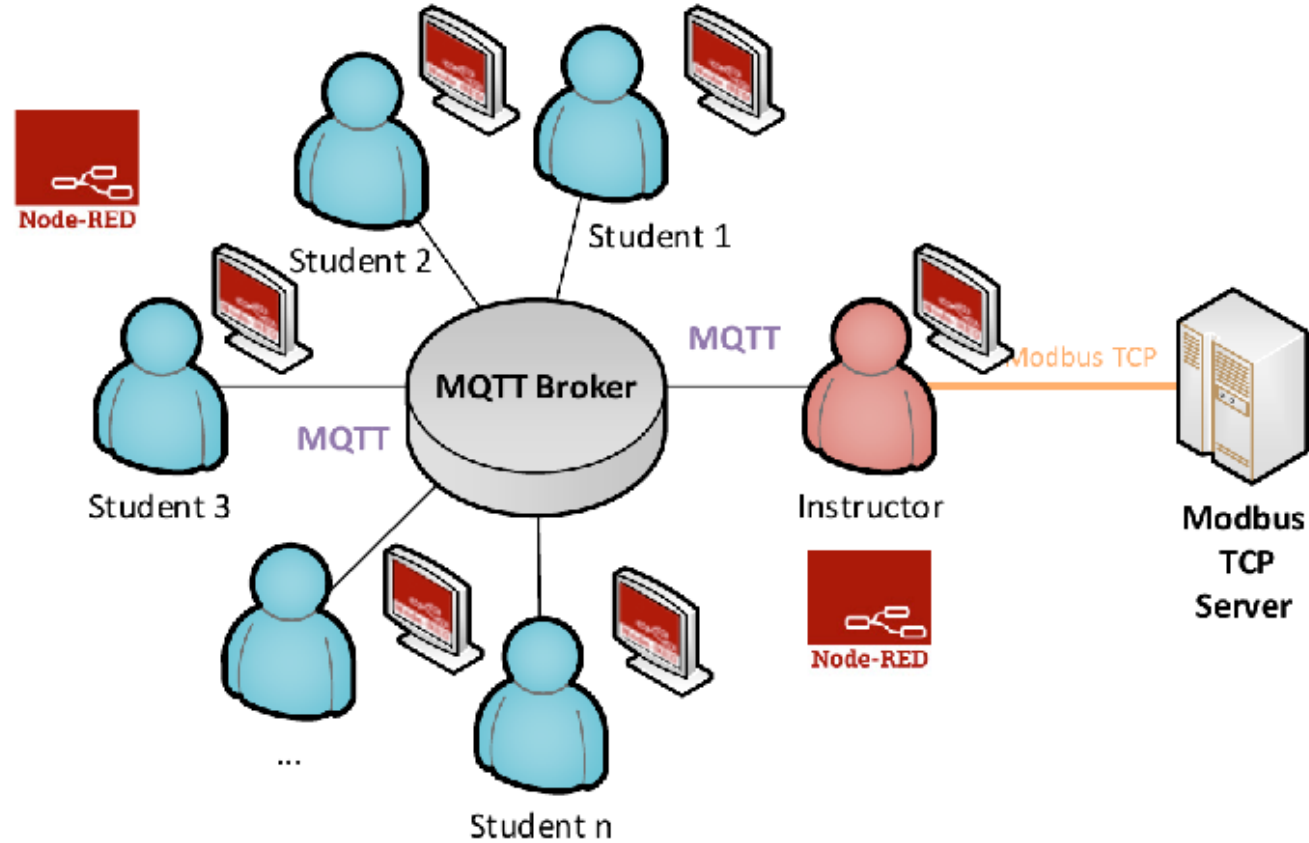
4-20mA Sensor Interfacing

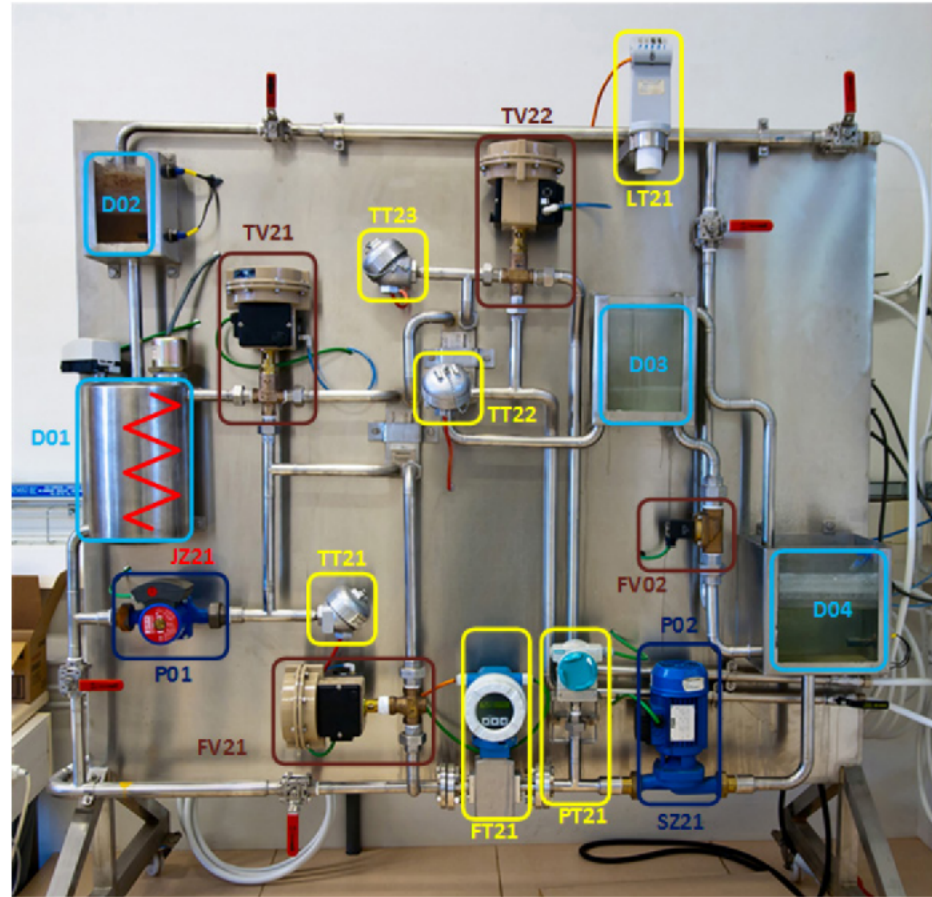
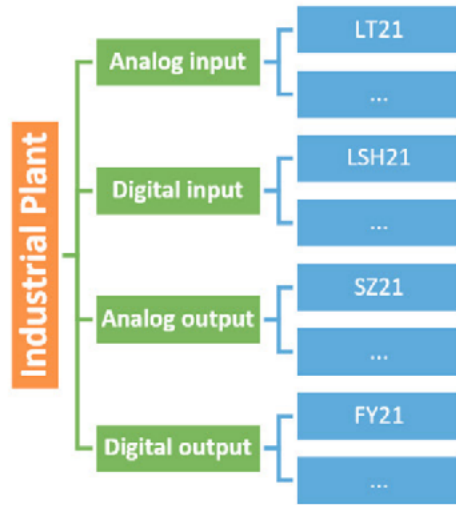


Source: <https://www.loullingen.lu/projekte/logo/index.php>

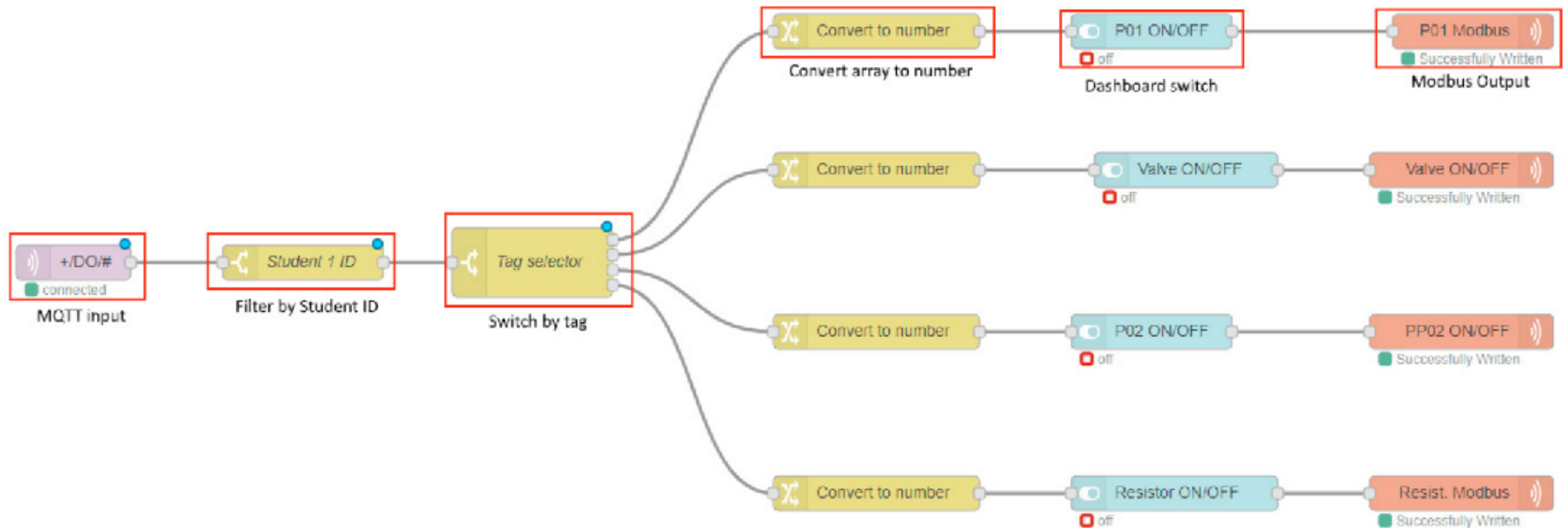
Paper: Development of a Remote Industrial Laboratory for Automatic Control based on Node-RED (2020)

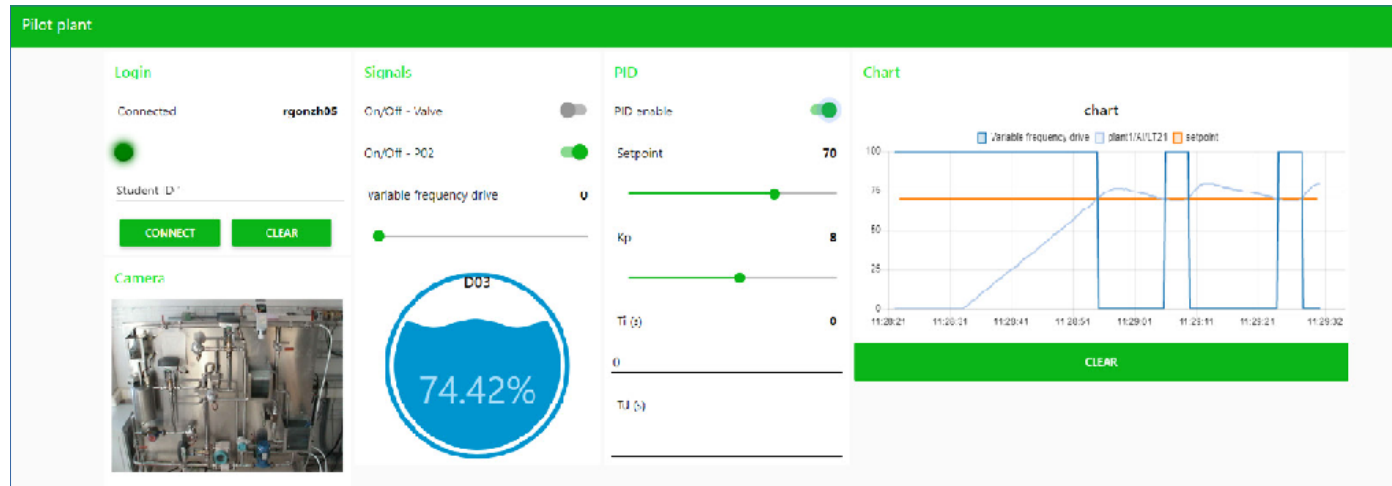
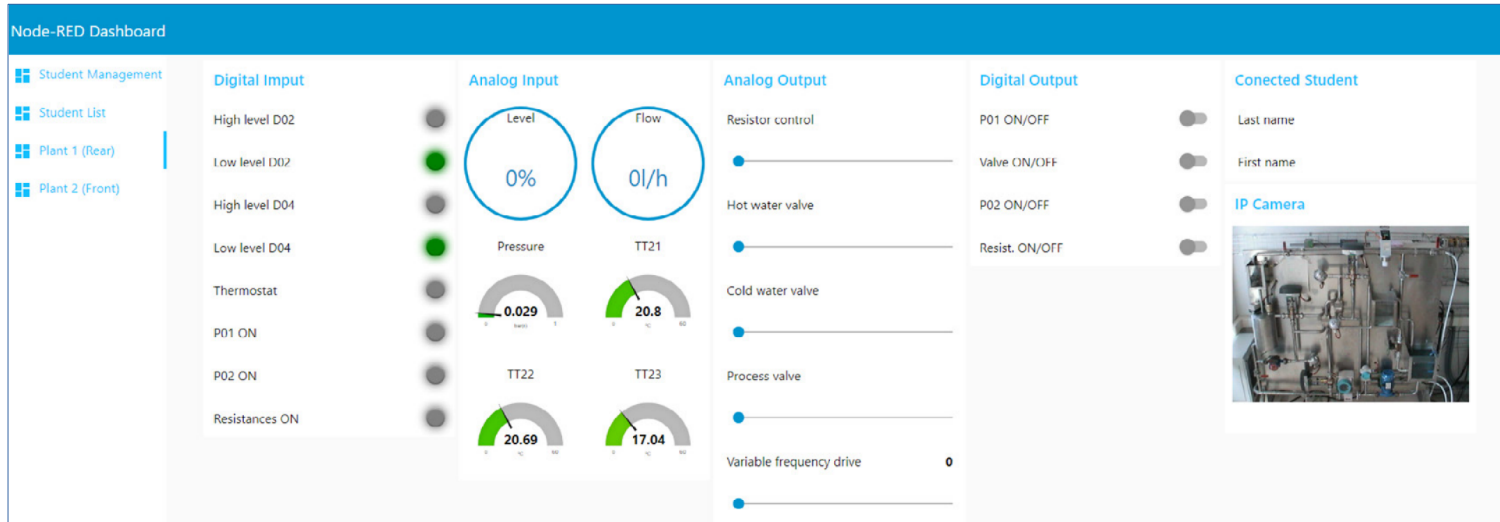
- Implements an automation control lab for remote access using Node-RED, MQTT, Modbus TCP.
- Node-RED flows (clients) ↔ an MQTT broker ↔ a Modbus TCP server.
- MQTT broker: Eclipse Mosquitto (open source) or Schneider Electric iPC Box IIoT (industrial PC designed for edge computing)
- PLC: Schneider Electric Modicon M340 + Modbus TCP protocol
- Node-RED module for the Modbus TCP used to communicate with the PLC: "node-red-contrib-modbustcp"





Paper: Development of a Remote Industrial Laboratory for Automatic Control based on Node-RED (2020)



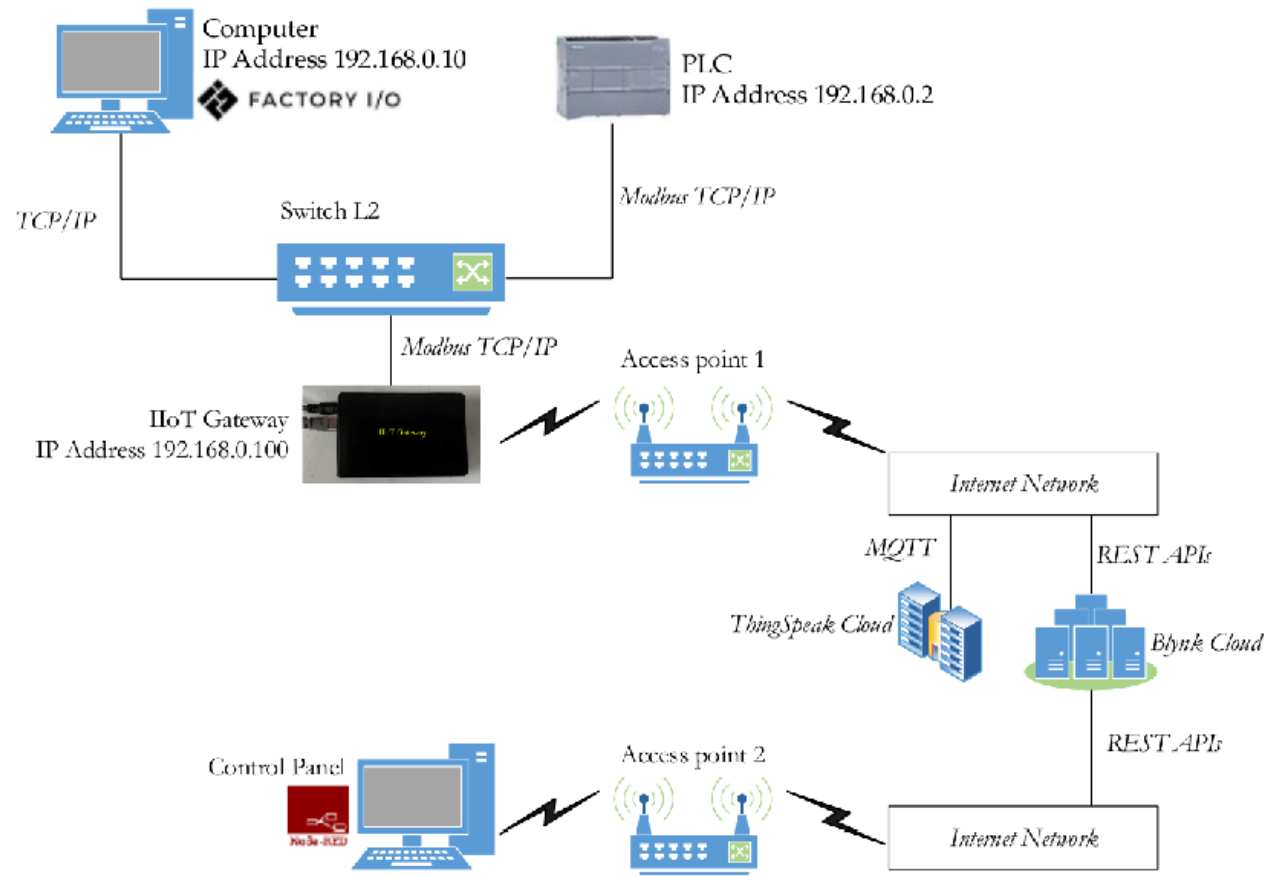


Paper: “*Cost-Effective IIoT Gateway Development Using ESP32 for Industrial Applications*” (2024)

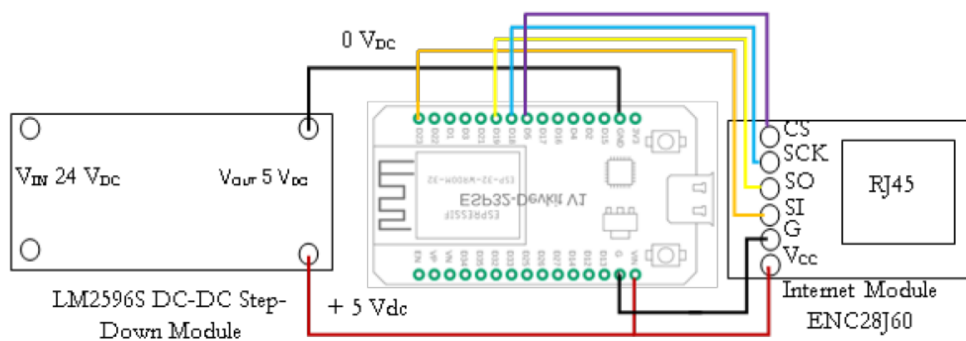
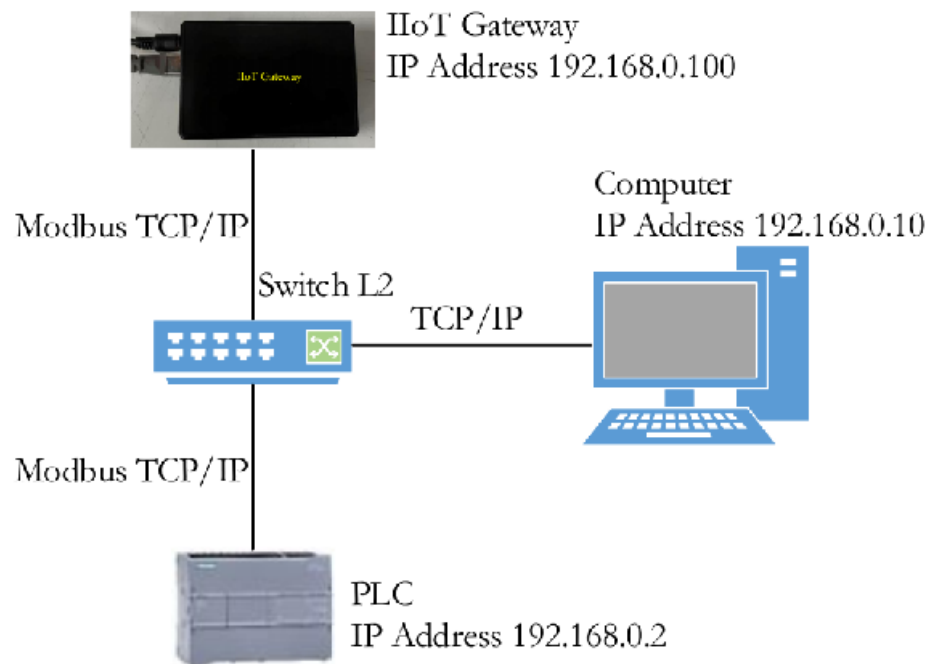
- Implements an ESP32-based IoT gateway instead of Siemens IOT2050 IIoT Gateway.
 - LM2596S DC-DC Step-Down module (24V -> 5V)
 - Ethernet module for ESP32: ENC28J60 (SPI interface, 10/100 Mbps)
 - ESP32 as Modbus TCP server and PLC as Modbus TCP client
- PLC: Siemens S7-1200 12144C AC/DC/RLY
- Communication Protocol: Modbus TCP/IP
- Cloud connection (e.g. ThingSpeak or Blink Cloud): MQTT, REST API
- Performance Measurement: Read/write 16-bit data to the PLC via Modbus TCP

Source: <https://engj.org/index.php/ej/article/view/4584>

Related Work: ESP32 as an IIoT Gateway



Related Work: ESP32 as an IoT Gateway



References: Related Work / Further Readings

- **Development of a Remote Industrial Laboratory for Automatic Control based on Node-RED**
=> <https://www.sciencedirect.com/science/article/pii/S2405896320323478>
- **Implementation of Industrial IoT Integration Using Node-RED and PLC on Cascade Control Level and Flow Plant**
Implementation of Industrial IoT Integration Using Node-RED and PLC on Cascade Control Level and Flow Plant (2025)
=> <https://jppipa.unram.ac.id/index.php/jppipa/article/view/11623>
- **Using Node-RED and MQTT with PLCnext Technology**
=> <https://www.realpars.com/blog/node-red-mqtt-plcnext>
- **Sugar Shack 4.0: Practical Demonstration of an IIoT-Based Event-Driven Automation System**
=> <https://arxiv.org/abs/2510.15708>

References: Related Work / Further Readings

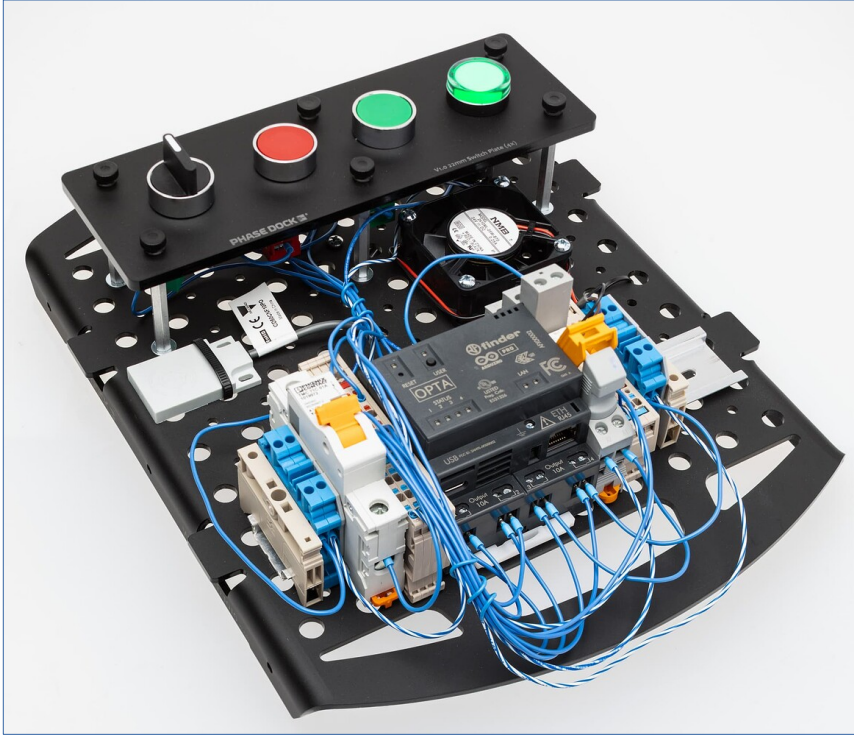
- **The Future of Intelligent Industrial Systems: PLC, Node-RED, and IoT / IIoT**
=> <https://ates.alayen.edu.iq/home/vol2/iss2/5/>
- **An Open-Source Supervisory Control and Data Acquisition Architecture for Photovoltaic System Monitoring Using ESP32, Banana Pi M4, and Node-RED**
=> <https://www.mdpi.com/1996-1073/17/10/2295>
- **Design and Implementation of Node-Red Based Open-Source SCADA Architecture for a Hybrid Power System**
=> <https://www.mdpi.com/1996-1073/16/5/2092>
- **An Internet of Things—Supervisory Control and Data Acquisition (IoT-SCADA) Architecture for Photovoltaic System Monitoring, Control, and Inspection in Real Time**
=> <https://www.mdpi.com/2079-9292/14/1/42>

References: Related Work / Further Readings

- **Supervisory Control and Data Acquisition Approach in Node-RED: Application and Discussions**
=> <https://www.mdpi.com/2624-831X/1/1/5>
- **An Open-Source Supervisory Control and Data Acquisition Architecture for Photovoltaic System Monitoring Using ESP32, Banana Pi M4, and Node-RED**
=> <https://www.mdpi.com/1996-1073/17/10/2295>
- **Design and Implementation of Node-Red Based Open-Source SCADA Architecture for a Hybrid Power System**
=> <https://www.mdpi.com/1996-1073/16/5/2092>
- **Home Assistant – LOGO! V8.4 MQTT Topics**
=> <https://community.home-assistant.io/t/siemens-logo-8-4-mqtt/747049>
-

References: Related Work / Further Readings

- **Smart Automation for Residential Spaces with PLC-ESP32 Architecture**
=> <https://www.mdpi.com/2673-4591/115/1/7>
- **The PLC as a Smart Service in Industry 4.0 Production Systems**
=> <https://www.mdpi.com/2076-3417/9/18/3815>
- **Development of a Remote Industrial Laboratory for Automatic Control based on Node-RED**
=> <https://www.sciencedirect.com/science/article/pii/S2405896320323478>
- **RedPLC: Node-Red nodes for the realization of Soft-PLC with Ladder-Logic**
=> <https://github.com/redplc/redplc>
=> <https://flows.nodered.org/node/@redplc/node-red-redplc>



PLC4training - Logo8-Trainingsboard for Siemens LOGO 8



Training Board for

- Logo! 8 12/24 RCE (6ED1052-1MD00-0BA8)
- Logo! DM8 12/24R (6ED1055-1MB00-0BA2)
- Extension modules: DM8 12/24R, AM2 or AM2 AQ

PLC4training - Logo8-Trainingsboard for Siemens LOGO 8

2x20-pin Interface connector



- 8x push-button/latching switches
- 4x analog sensors (0-10 Volt)
- 1x pulse generator (0-1000 Hz)
- 4x LEDs

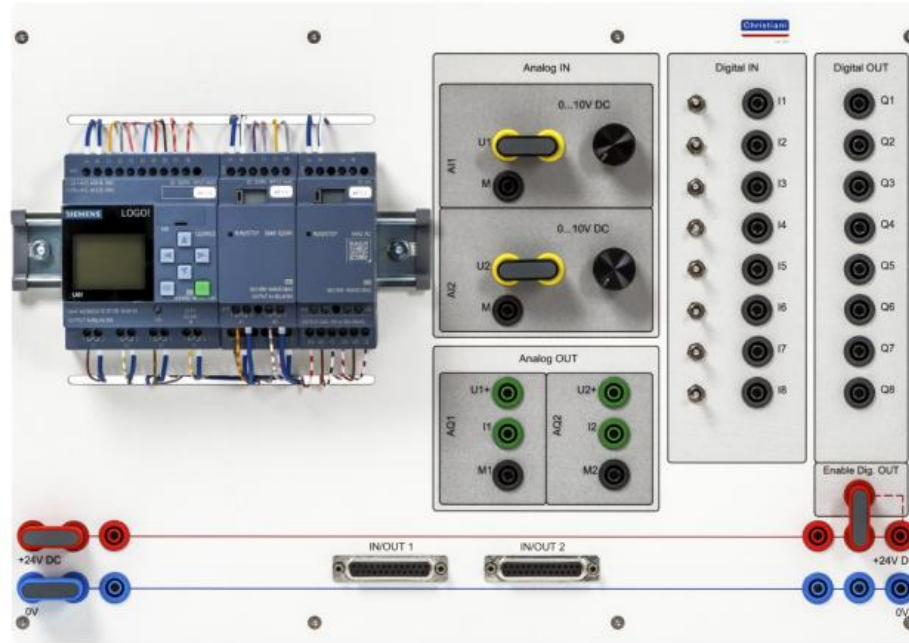
PLC4training - Logo8-Trainingsboard for Siemens LOGO 8



LOGO8 DM8



Training Board Logo! 8



LOGO! 8 12/24RCE

- 8 digital inputs (4 analog usable)
- 4 digital outputs
- Relay output

LOGO! DM8 12/24R expansion module

- Power supply Inputs: 12/24 V DC
- Outputs: Relay
- 4 DE (digital inputs)
- 4 DA (digital outputs)

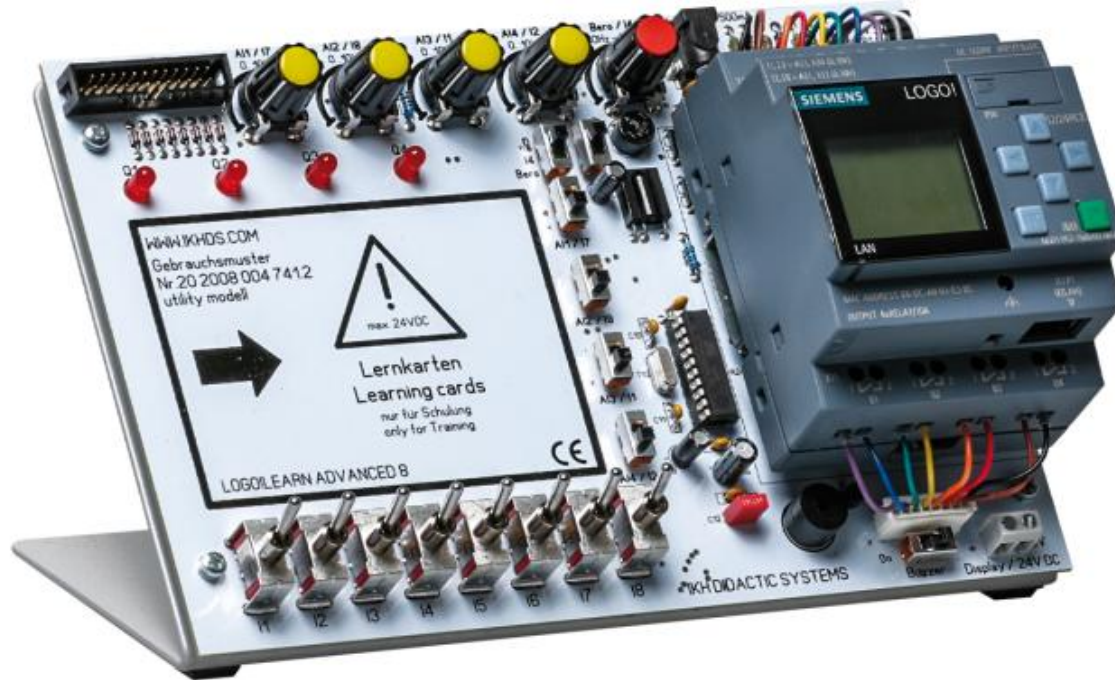
LOGO! LOGO! AM2 AQ expansion module

- Power supply: 24 V DC
- 2 AA (analogue outputs)
- Output ranges: 0 - 10 V
- Output ranges: 0/4 - 20 mA

Source:

<https://www.christiani-international.com/Vocational-Training-University/Automation-Technology-Electronics/Control-Engineering/Learning-Board-Logo-8.html>

LOGO!Learn Advanced



- Supply voltage: 12 - 24 V DC
- 8x push buttons / latching switches
- 4x potentiometers (analog voltage, up to 10V)
- 4x slide switches (analog input vs. digital inputs)